Customer Segmentation via Unsupervised Clustering

RFM Analysis and K-Means for Actionable Marketing Segments

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Agenda

1. Business Context
2. Dataset & EDA
3. Clustering Analysis
4. Segment Profiling
5. Marketing Strategy
6. Conclusions

Why Segment Customers?

The Problem

- One-size-fits-all marketing wastes **40% of budget**
- VIP customers need **different treatment** than dormant ones
- Without segmentation, retention efforts are **untargeted**

Expected Outcomes

Goal	Target
Segments	4–6
Silhouette	> 0.25
Coverage	100%
Actionability	High

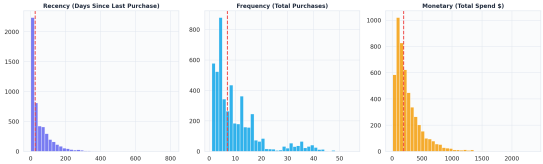
Our Approach

Apply **unsupervised clustering** on RFM and behavioral features to discover natural customer groups, then map each to a tailored marketing strategy.

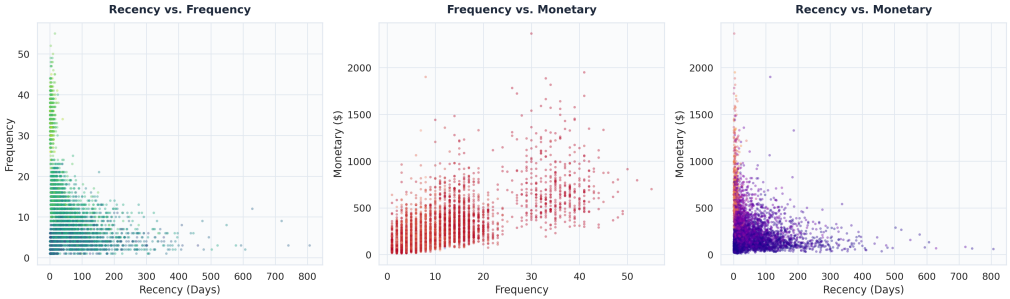
E-Commerce Customer Data

Dataset

- **5,000** e-commerce customers
- **10** features: RFM, tenure, AOV, categories, demographics
- Recency: 1–290 days
- Frequency: 1–45 purchases
- Monetary: $10–$5,000+



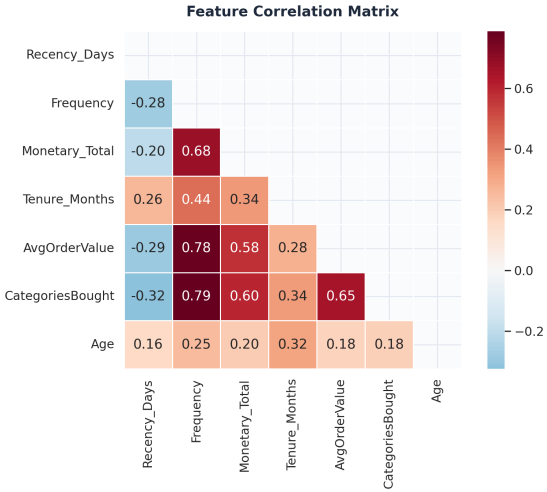
RFM Scatter Analysis

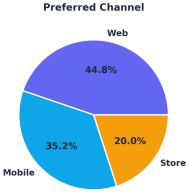


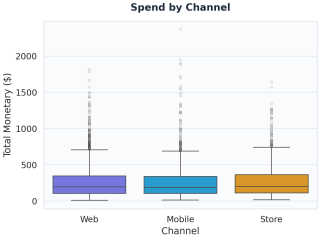
Observation

Clear clusters emerge in RFM space: high-frequency/high-monetary customers (top-right) vs. low-engagement/low-spend (bottom-left). These natural groupings motivate formal clustering.

Correlation & Channel Analysis

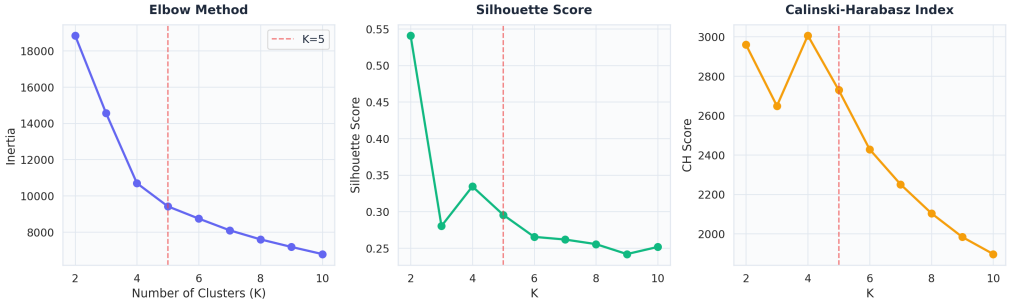






Insights

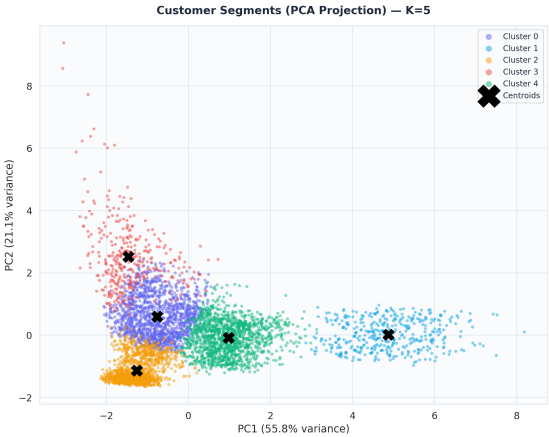
Optimal K Selection

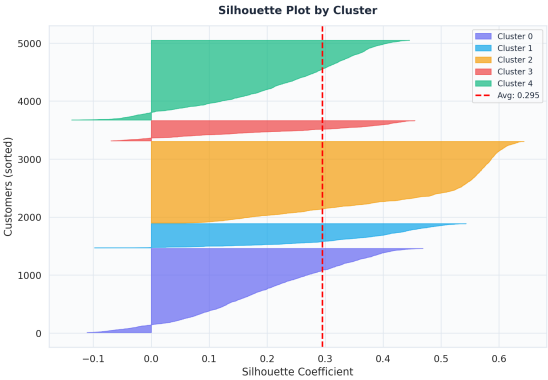


Decision: K = 5

The elbow method shows diminishing returns after K=5. Silhouette score peaks in the 4–6 range. We select **K=5** for interpretability and balanced segment sizes.

PCA Visualization & Silhouette Analysis

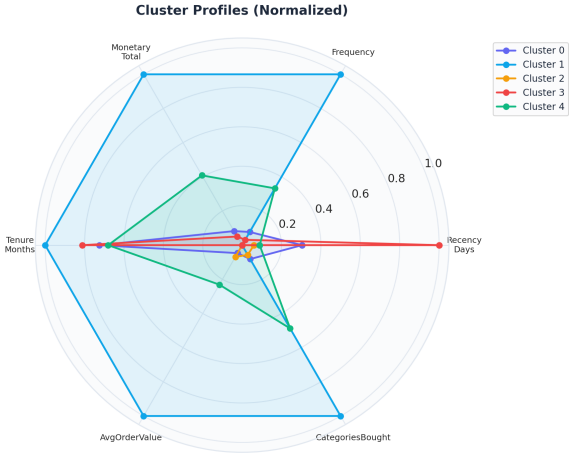




Validation

PCA projection shows **clear separation** between clusters. Silhouette plot confirms most

Cluster Radar Profiles

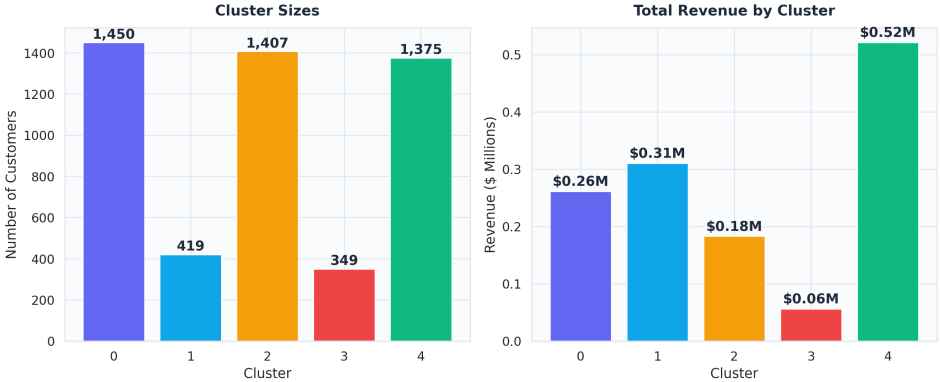


Segment Archetypes

Each cluster has a **distinct behavioral fingerprint**:

- High frequency + low recency = **VIP**
- High recency + moderate freq = **At-Risk**
- Low everything = **Dormant**
- Recent + low freq = **New**
- Balanced = **Regular**

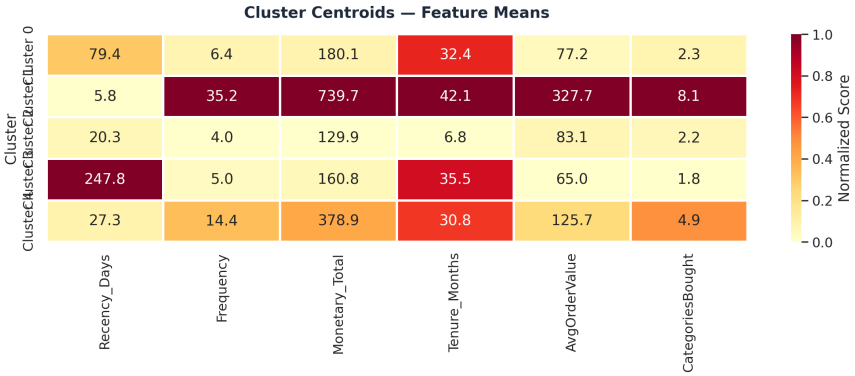
Size, Revenue & Feature Distributions



Revenue Concentration

A small number of VIP customers typically generate a **disproportionate share** of total revenue — reinforcing the importance of segment-specific retention strategies.

Segment Centroids Heatmap



Interpretation

Each row represents a cluster's average feature values. The color intensity reveals which features define each segment — enabling clear, data-driven marketing personas.

Segment-Specific Actions

High-Value Segments

VIP Champions Exclusive rewards, personal account manager, early access to new products

Loyal Regulars Loyalty points, cross-sell recommendations, referral bonuses

At-Risk & Recovery

Promising New Onboarding emails, first-purchase incentives, product education

At-Risk Win-back campaigns, satisfaction surveys, retention offers

Dormant Deep discounts, reactivation emails, feedback requests

Summary & Future Work

Achievements

- **5 distinct segments** discovered
- Silhouette score: **0.30**
- Clear PCA separation
- Actionable marketing strategies
- Revenue attribution per segment

Future Work

1. DBSCAN/HDBSCAN for outliers
2. Deep clustering (autoencoders)
3. CLV prediction per segment
4. A/B test segment strategies
5. Real-time segmentation pipeline

Thank You

Questions & Discussion

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